Anga Sai Girish

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B.Tech - Computer Science and Engineering
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Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. Major	Indian Institute of Technology, Guwahati	7.19 (Current)	2022-Present
Senior Secondary	Board of Intermediate Education, Andhra Pradesh	93.2%	2022
Secondary	Board of Secondary Education, Andhra Pradesh	100%	2020

Projects

- **SafeScenePro: Systematic Discovery and Mitigation of AV Perception Failures** *Jun. 2025 - Present*
  Personal Research Project Private GitHub - Access upon request
  - Developed a **physics-based** edge case generator in **PyTorch**, implementing volumetric fog rendering (Beer–Lambert law), atmospheric light scattering, chromatic aberration, and 7-level multi-exposure fusion to process **662** scenarios.
  - Built a multi-architecture evaluation pipeline for **YOLOv5**, **YOLOv8**, and **Faster R-CNN**, with custom IoU-based object tracking that achieved **95.9% selectivity** from 16,000 candidates using **CLAHE** and histogram equalization.
  - Integrated **12,607 AV disengagement reports** (via pandas/CSV parsing) with **NHTSA crash data** to validate scenarios, implementing automated categorization across four failure modes using **OpenCV** brightness and contrast analysis.
  - Created an **end-to-end system** that processed **Cityscapes/KITTI datasets** through the weather physics engine, generating 500 synthetic training samples with YOLO-format annotations and achieving **99.5% mAP** on synthetic validation.
- **DocuMind – Intelligent Document Assistant (RAG)** *May 2025 - Aug. 2025*
  Personal Project bit.ly/Qa_RAG
  - Engineered an intelligent document Q&A system with a GPU-accelerated RAG using **PyTorch CUDA**, **Faiss**, **FastAPI**, **PostgreSQL** and **React**, achieving **sub-100ms** embedding generation on RTX GPUs with automatic CPU fallback.
  - Developed a multi-format document pipeline (pdf, docx, txt, md, html) with **PyPDF2**, **python-docx** and **BeautifulSoup4**, integrating **LLMs** (**GGUF**, 4-bit **Transformers**, **OpenAI/Groq**) and persistent **Faiss** indexing.
  - Deployed containerized microservices with **Docker**, providing real-time progress tracking through **WebSockets**, **FastAPI BackgroundTasks**, and a responsive **React/Zustand** frontend styled with **Tailwind CSS** and **Framer Motion**.
  - Built an analytics dashboard in **React/Recharts** visualizing **8ms** vector search latency and **GPU vs CPU throughput (120 vs 8–10 docs/min)**, with memory optimizations (singleton model, batching, GC) and offline-ready **PWA** support.
- **Real-time Network Intrusion Detection System (NIDS)** *Aug. 2025 - Sep. 2025*
  Group Project bit.ly/Nids
  - Architected a real-time cybersecurity monitoring system handling **1,000-3,000 packets/sec** using a **multi-threaded Python architecture** and **Berkeley Packet Filter (BPF)** optimization, achieving **60**% performance improvement.
  - Delivered <**200ms** alert response time by implementing **8 threat detection algorithms** (SYN Flood, Port Scanning, ARP Spoofing, ICMP Flood, LAND, etc.) with configurable thresholds and geolocation automation.
  - Built a full-stack **security dashboard** with live **WebSocket** updates, **RESTful API** (4 endpoints), and interactive threat visualizations in **Plotly**, featuring **geographic IP tracking** and persistent **SQLite** for enterprise-grade threat analysis.
- **Traffic Severity Classification** *Feb. 2025 - Mar. 2025*
  Group project bit.ly/TSPredict
  - Built an end-to-end Traffic Severity **Classification pipeline**: Engineered **15**+ highly informative new features from a dataset of **100,000**+ traffic incidents by processing temporal, weather, and road characteristics.
  - Trained and optimized **Random Forest, XGBoost, and LightGBM** models extensively, with the final **LightGBM** achieving **92.4**% accuracy and a **Quadratic Weighted Kappa** of **0.846**.
  - Performed **feature & error analysis**: Identified Duration_Min, Start_Lng, and Pressure as top predictors and revealed challenges predicting rare severity classes **1 and 4** (error rates **85%** and **100%**); benchmarked against an **LLM baseline**.

Achievements & Extracurricular Activities

- **Codeforces**,Achieved a maximum rating of **1681** (**Expert**) *Handle*
  - Codeforces Round 1044: Secured **Global Rank 653** among **16,000**+ participants.
  - Codeforces Round 1034: Secured **Global Rank 1121** among **15,000**+ participants.
- **Leetcode**, Secured **Global Rank 415** among **37,000**+ participants in Leetcode Biweekly Contest 162. *Handle*
- **Codechef**, Secured **Global Rank 24** in Codechef Starters 166. *Handle*
- **JEE Advanced**, Secured **All India Rank 1592** among 0.15 million candidates. *2022*
- **JEE Mains**, Secured **All India Rank 1135** among 1 million candidates. *2022*
- **AP EAPCET**, Secured **State-wide Rank 118** among 0.16 million candidates. *2022*
- **TS EAMCET**, Secured **State-wide Rank 317** among 0.12 million candidates. *2022*
- **Cricket League**, Achieved a **Man of the Match** award in the Srikakulam Cricket League. *2023*
- **NSS**, Contributed to community outreach as part of the **National Service Scheme**, IIT Guwahati. *2024*

Technical Skills

- **Languages:** C, C++, Python
- **ML/AI:** Transformers, LLMs, Faiss, PyTorch*, CUDA*
- **Backend:** Node.js, FastAPI, Express.js, WebSockets
- **Database Management:** MongoDB, PostgreSQL, MySQL
- **Tools:** Docker, Git, WebSockets, pandas, Streamlit*
- **Frontend:** React.js*, Tailwind CSS

** Elementary proficiency*

Key Courses Taken

- **Computer Science**: Data Structures and Algorithms, Machine Learning, Software Engineering, High Performance Computing, Database Management Systems, Computer Architecture and Organization, Implementation of Programming Languages Lab, Computer Networks, Operating Systems.
- **Mathematics**: Probability Theory and Random Processes, Elementary Number Theory and Algebra, Optimization.